Equations give

Substituting into gives

The electrical energy obtainable from this kinetic energy is

Obtaining requires the iceberg to absorb of heat. Melting it absorbs a total of

Hence the total electrical energy obtainable is

Substituting and into gives

Substituting the numerical data gives

Scoring (points): ; ; ; ; ;
; each .

Method
At ambient pressure , let moles of air undergo the process as all the ice melts. In step , volume changes by and temperature by . The air releases the heat

On the right, term one is atmospheric work on the system; term two is the internal energy change. The equation of state gives